

Exercises Chapter 1: The Time Value of Money**Basic Exercises****The Timeline**

- 1) You have just taken out a five-year loan from a bank to buy an engagement ring. The ring costs \$7,000. You plan to put down \$1,000 and borrow \$6,000. You will need to make annual payments of \$1,250 at the end of each year. Show the timeline of the loan from your perspective. How would the timeline differ if you created it from the bank's perspective? (BDM 4.1)

The Three Rules of Time Travel

- 2) Calculate the future value of \$7,000 in
 - a. Four years at an interest rate of 8% per year.
 - b. Eight years at an interest rate of 8% per year.
 - c. Four years at an interest rate of 16% per year.
 - d. Why is the amount of interest earned in part (a) less than half the amount of interest earned in part (b)? (BDM 4.3)
- 3) You are thinking of retiring. Your retirement plan will pay you either \$250,000 immediately on retirement or \$350,000 five years after the date of your retirement. Which alternative should you choose if the interest rate is: (BDM 4.9)
 - a. 0% per year?
 - b. 8% per year?
 - c. 20% per year?
- 4) Your grandfather put some money in an account for you on the day you were born. You are now 18 years old and are allowed to withdraw the money for the first time. The account currently has \$3,247 in it and pays a 6% interest rate. (BDM 4.10)
 - a. How much money would be in the account if you left the money there until your 25th birthday?
 - b. What if you left the money until your 65th birthday?
 - c. How much money did your grandfather originally put in the account?

Valuing a Stream of Cash Flows

- 5) You have a loan outstanding. It requires making six annual payments at the end of the next six years of \$9,000 each. Your bank has offered to restructure the loan so that instead of making the six payments as originally agreed, you will make only one final payment at the end of the loan in six years. If the interest rate on the loan is 9.85%, what final payment will the bank require you to make so that it is indifferent between the two forms of payment? (BDM 4.13)

Calculating the Net Present Value

- 6) Marian Plunket owns her own business and is considering an investment. If she undertakes the investment, it will pay \$8,000 at the end of each of the next three years. The opportunity requires an initial investment of \$2,000 plus an additional investment at the end of the second year of \$10,000. What is the NPV of this opportunity if the interest rate is 10% per year? Should Marian take it? (BDM 4.15)

Perpetuities and Annuities

- 7) Your buddy in mechanical engineering has invented a money machine. The main drawback of the machine is that it is slow. It takes one year to manufacture \$100. However, once built, the machine will last forever and will require no maintenance. The machine can be built immediately, but it will cost \$1,000 to build. Your buddy wants to know if he should invest the money to construct it. If the interest rate is 9.5% per year, what should your buddy do? (BDM 4.16)
- 8) How would your answer to Problem 7 change if the machine takes one year to build? (BDM 4.17)
- 9) You are running a hot Internet company. Analysts predict that its earnings will grow at 30% per year for the next five years. After that, as competition increases, earnings growth is expected to slow to 5% per year and continue at that level forever. Your company has just announced earnings of \$1,000,000. What is the present value of all future earnings if the interest rate is 10%? (Assume all cash flows occur at the end of the year.) (BDM 4.29)

Solving for the Cash Payments

- 10) You are saving for retirement. To live comfortably, you decide you will need to save \$2 million by the time you are 65. Today is your 30th birthday, and you decide, starting today and continuing on every birthday up to and including your 65th birthday, that you will put the same amount into a savings account. If the interest rate is 5%, how much must you set aside each year to make sure that you will have \$2 million in the account on your 65th birthday? (BDM 4.42)

The Internal Rate of Return

- 11) You have an investment opportunity that requires an initial investment of \$6,500 today and will pay \$7,500 in one year. What is the IRR of this opportunity? (BDM 4.47)
- 12) You are shopping for a car and read the following advertisement in the newspaper: "Own a new Spitfire! No money down. Four annual payments of just \$15,000." You have shopped around and know that you can buy a Spitfire for cash for \$45,000. What is the interest rate the dealer is advertising (what is the IRR of the loan in the advertisement)? Assume that you must make the annual payments at the end of each year. (BDM 4.49)
- 13) Nielson Motors is considering an opportunity that requires an investment of \$1,000,000 today and will provide \$250,000 one year from now, \$450,000 two years from now, and \$650,000 three years from now. The internal rate of return of this project is closest to: (BDM test bank 5)
- A) 10.2%.
 - B) 12.2%.
 - C) 14.2%.
 - D) 16.2%.

Exercises Chapter 2: Investment Decision Rules**Basic Exercises****NPV and Stand-Alone Projects**

- 1) Your brother wants to borrow \$10,250 from you. He has offered to pay you back \$12,500 in a year. If the cost of capital of this investment opportunity is 8%, what is its NPV? Should you undertake the investment opportunity? Calculate the IRR and use it to determine the maximum deviation allowable in the cost of capital estimate to leave the decision unchanged. (BDM 7.1)
- 2) Bill Clinton reportedly was paid \$10 million to write his book My Life. Suppose the book took three years to write. In the time he spent writing, Clinton could have been paid to make speeches. Given his popularity, assume that he could earn \$8 million per year (paid at the end of the year) speaking instead of writing. Assume his cost of capital is 10% per year. (BDM 7.5)
 - a. What is the NPV of agreeing to write the book (ignoring any royalty payments)?
 - b. Assume that, once the book is finished, it is expected to generate royalties of \$5 million in the first year (paid at the end of the year) and these royalties are expected to decrease at a rate of 30% per year in perpetuity. What is the NPV of the book with the royalty payments?

The Internal Rate of Return Rule

- 3) You are considering an investment in a clothes distributor. The company needs \$105,000 today and expects to repay you \$125,000 in a year from now. What is the IRR of this investment opportunity? Given the riskiness of the investment opportunity, your cost of capital is 18%. What does the IRR rule say about whether you should invest? (BDM 7.9)
- 4) Innovation Company is thinking about marketing a new software product. Upfront costs to market and develop the product are \$5.3 million. The product is expected to generate profits of \$1.4 million per year for 10 years. The company will have to provide product support expected to cost \$94,000 per year in perpetuity. Assume all profits and expenses occur at the end of the year. (BDM 7.14)
 - a. What is the NPV of this investment if the cost of capital is 5.97%? Should the firm undertake the project? Repeat the analysis for discount rates of 1.19% and 20.58%.
 - b. Deleted.
 - c. Can the IRR rule be used to evaluate this investment? Explain.

- 5) You have 3 projects with the following cash flows:

Year	0	1	2	3	4
Project 1	-152	19	41	58	81
Project 2	-823	0	0	6995	-6496
Project 3	21	38	62	81	-244

For which of these projects is the IRR rule reliable? (BDM 7.15)

The Payback Rule

- 6) You are considering making a movie. The movie is expected to cost \$10.7 million upfront and take a year to make. After that, it is expected to make \$4.3 million in the first year it is released and \$1.7 million per year for the following four years. What is the payback period of this investment? If you require a payback period of two years, will you make the movie? Does the movie have positive NPV if the cost of capital is 10.1%? According to the NPV rule, should you make this movie? (BDM 7.22)

Choosing between Projects

- 7) Consider two investment projects, both of which require an upfront investment of \$11 million and pay a constant positive amount each year for the next 10 years. Under what conditions can you rank these projects by comparing their IRRs? (BDM 7.28)
- 8) You are deciding between two mutually exclusive investment opportunities. Both require the same initial investment of \$10 million. Investment A will generate \$2 million per year (starting at the end of the first year) in perpetuity. Investment B will generate \$1.5 million at the end of the first year and its revenues will grow at 2% per year for every year after that. (BDM 7.23)
- Which investment has the higher IRR?
 - Which investment has the higher NPV when the cost of capital is 7%?
 - In this case, for what values of the cost of capital does picking the higher IRR give the correct answer as to which investment is the best opportunity?

Project Selection with Resource Constraints

- 9) Natasha's Flowers, a local florist, purchases fresh flowers each day at the local flower market. The buyer has a budget of \$950 per day to spend. Different flowers have different profit margins, and also a maximum amount the shop can sell. Based on past experience, the shop has estimated the following NPV of purchasing each type:

	NPV per Bunch	Cost per Bunch	Max. Bunches
Roses	\$1	\$25	25
Lilies	10	22	10
Pansies	6	31	10
Orchids	21	84	5

What combination of flowers should the shop purchase each day? (BDM 7.31)

- 10) Kaimalino Properties (KP) is evaluating six real estate investments. Management plans to buy the properties today and sell them five years from today. The following table summarizes the initial cost and the expected sale price for each property, as well as the appropriate discount rate based on the risk of each venture.

Project	Cost Today	Discount Rate	Expected Sale Price in Year 5
Mountain Ridge	\$ 3,000,000	15%	\$18,000,000
Ocean Park Estates	15,000,000	15%	75,500,000
Lakeview	9,000,000	15%	50,000,000
Seabreeze	6,000,000	8%	35,500,000
Green Hills	3,000,000	8%	10,000,000
West Ranch	9,000,000	8%	46,500,000

KP has a total capital budget of \$18,000,000 to invest in properties. (BDM 7.33)

- What is the IRR of each investment?
- What is the NPV of each investment?
- Given its budget of \$18,000,000, which properties should KP choose?
- Explain why the profitability index method could not be used if KP's budget were \$12,000,000 instead. Which properties should KP choose in this case?

Exercises Chapter 3: Bonds**Basic Exercises****Bond Cash Flows, Prices, and Yields**

- 1) A 15-year bond with a face value of \$1,000 has a coupon rate of 4.5%, with semiannual payments. (BDM 6.1)
 - a. What is the coupon payment for this bond?
 - b. Draw the cash flows for the bond on a timeline.

Dynamic Behavior of Bond Prices

- 2) The prices of several bonds with face values of \$1,000 are summarized in the following table: (BDM 6.8)

Bond	A	B	C	D
Price	\$936.57	\$1,095.48	\$1,170.97	\$1,000.00

For each bond, state whether it trades at a discount, at par, or at a premium.

- 3) Explain why the yield of a bond that trades at a discount exceeds the bond's coupon rate. (BDM 6.9)
- 4) Suppose you purchase a 10-year bond with 4% annual coupons. You hold the bond for four years, and sell it immediately after receiving the fourth coupon. If the bond's yield to maturity was 3.75% when you purchased and sold the bond, (BDM 6.12)
 - a. What cash flows will you pay and receive from your investment in the bond per \$100 face value?
 - b. What is the internal rate of return of your investment?
- 5) Suppose you purchase a 30-year, zero-coupon bond with a yield to maturity of 6% and a face value of \$100. You hold the bond for five years before selling it. (BDM 6.14)
 - a. If the bond's yield to maturity is 6% when you sell it, what is the internal rate of return of your investment?
 - b. If the bond's yield to maturity is 7% when you sell it, what is the internal rate of return of your investment?
 - c. If the bond's yield to maturity is 5% when you sell it, what is the internal rate of return of your investment?
 - d. Even if a bond has no chance of default, is your investment risk free if you plan to sell it before it matures? Explain.

Investing in Bonds

- 6) A German federal bond (Bundesanleihe) with an annual 4% coupon, a remaining term of 5 years and a nominal value of EUR 100 is traded on the market. The bond is redeemed at 100% at maturity. Currently, the market return for German government bonds is 2.0%. (InvFin slides_2/2015)
- What is the market value of the 4% federal bond?
 - An investor buys the German government bond at the market price and wants to sell it after exactly three years, i.e., right after the coupon payment. The market return on German government bonds has meanwhile risen to 3.5%. At what price will he sell the bond? What is the internal rate of return (IRR) on his invested capital if he sells the bond after three years?

Corporate Bonds

- 7) The following table summarizes the yields to maturity on several one-year, zero-coupon securities: (BDM 6.29)

Security	Yield (%)
Treasury	2.940
AAA corporate	3.633
BBB corporate	6.259
B corporate	8.061

- What is the price of a one-year, zero-coupon corporate bond with a face value of \$100 and with a AAA rating?
- What is the credit spread on AAA-rated corporate bonds?
- What is the credit spread on B-rated corporate bonds?
- How does the credit spread change with the bond rating? Why?

- 8) HMK Enterprises would like to raise \$14 million to invest in capital expenditures. The company plans to issue five-year bonds with a face value of \$1,000 and a coupon rate of 4% (annual payments). The following table summarizes the yield to maturity for five-year (annual-pay) coupon corporate bonds of various ratings: (BDM 6.31)

Rating	AAA	AA	A	BBB	BB
YTM	3.7%	3.9%	4%	4.7%	6.5%

- Assuming the bonds will be rated AA, what will the price of the bonds be?
- How much total principal amount of these bonds must HMK issue to raise \$14 million today, assuming the bonds are AA rated? (Because HMK cannot issue a fraction of a bond, assume that all fractions are rounded to the nearest whole number.)
- What must the rating of the bonds be for them to sell at par?
- Suppose that when the bonds are issued, the price of each bond is \$896.11. What is the likely rating of the bonds? Are they junk bonds?

Sovereign Bonds

- 9) Suppose the yield on German government bonds is 1.2%, while the yield on Spanish government bonds is 7%. Both bonds are denominated in euros. Which country do investors believe is more likely to default? How can you tell? (BDM 6.35)

Exercises Chapter 4: Stocks**Basic Exercises****Dividend-Discount Model (DDM)**

- 1) Assume Evco, Inc., has a current price of \$45 and will pay a \$2.05 dividend in one year, and its equity cost of capital is 16%. What price must you expect it to sell for right after paying the dividend in one year in order to justify its current price? (BDM 9.1)
- 2) Anle Corporation has a current price of \$17, is expected to pay a dividend of \$2 in one year, and its expected price right after paying that dividend is \$18. (BDM 9.2)
 - a. What is Anle's expected dividend yield?
 - b. What is Anle's expected capital gain rate?
 - c. What is Anle's equity cost of capital?
- 3) Suppose Acap Corporation will pay a dividend of \$2.84 per share at the end of this year and \$3.01 per share next year. You expect Acap's stock price to be \$50.68 in two years. If Acap's equity cost of capital is 8.3%: (BDM 9.3)
 - a. What price would you be willing to pay for a share of Acap stock today, if you planned to hold the stock for two years?
 - b. Suppose instead you plan to hold the stock for one year. What price would you expect to be able to sell a share of Acap stock for in one year?
 - c. Given your answer in part (b), what price would you be willing to pay for a share of Acap stock today, if you planned to hold the stock for one year? How does this compare to your answer in part (a)?

Applying the DDM

- 4) NoGrowth Corporation currently pays a dividend of \$1.36 per year, and it will continue to pay this dividend forever. What is the price per share if its equity cost of capital is 15% per year? (BDM 9.5)
- 5) Summit Systems will pay a dividend of \$1.50 this year. If you expect Summit's dividend to grow by 6% per year, what is its price per share if its equity cost of capital is 11%? (BDM 9.6)

- 6) DFB, Inc., expects earnings at the end of this year of \$4.19 per share, and it plans to pay a \$2.43 dividend at that time. DFB will retain \$1.76 per share of its earnings to reinvest in new projects with an expected return of 15.1% per year. Suppose DFB will maintain the same dividend payout rate, retention rate, and return on new investments in the future and will not change its number of outstanding shares. (BDM 9.10)
- What growth rate of earnings would you forecast for DFB?
 - If DFB's equity cost of capital is 12.2%, what price would you estimate for DFB stock today?
 - Suppose DFB instead paid a dividend of \$3.43 per share at the end of this year and retained only \$0.76 per share in earnings. If DFB maintains this higher payout rate in the future, what stock price would you estimate now? Should DFB raise its dividend?
- 7) Halliford Corporation expects to have earnings this coming year of \$3.15 per share. Halliford plans to retain all of its earnings for the next two years. For the subsequent two years, the firm will retain 53% of its earnings. It will then retain 20% of its earnings from that point onward. Each year, retained earnings will be invested in new projects with an expected return of 21.36% per year. Any earnings that are not retained will be paid out as dividends. Assume Halliford's share count remains constant and all earnings growth comes from the investment of retained earnings. If Halliford's equity cost of capital is 10.5%, what price would you estimate for Halliford stock? (BDM 9.15)

Market Efficiency

- 8) Apnex, Inc., is a biotechnology firm that is about to announce the results of its clinical trials of a potential new cancer drug. If the trials were successful, Apnex stock will be worth \$75 per share. If the trials were unsuccessful, Apnex stock will be worth \$18 per share. Suppose that the morning before the announcement is scheduled, Apnex shares are trading for \$52 per share. (BDM 9.33)
- Based on the current share price, what sort of expectations do investors seem to have about the success of the trials?
 - Suppose hedge fund manager Paul Kliner has hired several prominent research scientists to examine the public data on the drug and make their own assessment of the drug's promise. Would Kliner's fund be likely to profit by trading the stock in the hours prior to the announcement?
- 9) Geothermal Corp. just announced good news: Its earnings have increased by 20%. Most investors had anticipated an increase of 25%. Will Geothermal's stock price increase or decrease when the announcement is made? (BMM 7.133)

- 10) Identify and briefly describe the three forms of the Efficient Market Hypothesis. Explain the opportunity for abnormal returns in each form. (CorpFin17 D2 T2)

- 11) Please read the following text from the Financial Times.

99% of actively managed US equity funds underperform

Almost all US, global and EM (=Emerging Markets) funds have failed to beat their benchmark since 2006

The suitability of investing with asset managers that try to beat the market has been thrown into question by figures that show almost all US, global and emerging market funds have failed to outperform since 2006. The damning figures, provided by S&P Dow Jones, the index provider, will heap more pressure on active fund managers, which have come under scrutiny for charging investors high fees for poor performance. Academics and consumer groups have repeatedly attacked the industry for failing investors, and the UK financial regulator has launched an investigation into the workings of the asset management industry, the first of its kind, to examine fund fees and profitability. According to the analysis, 99 per cent of actively managed US equity funds sold in Europe have failed to beat the S&P 500 over the past 10 years, while only two in every 100 global equity funds have outperformed the S&P Global 1200 since 2006. Almost 97 per cent of emerging market funds have underperformed. Daniel Ung, director of research at S&P Dow Jones Indices, said: "The figures are startling."

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Explain potential reasons for the underperformance of actively managed funds. What does it say about the form of market efficiency prevailing in security markets? (CorpFin17 D2 T2)

Exercises Chapter 5: Capital Asset Pricing Model**Basic Exercises****Diversification**

- 1) The common stock of Leaning Tower of Pita, Inc., a restaurant chain, will generate the following payoffs to investors next year: (BMM 11.PP14)

State	Dividend	Stock Price
Boom	\$8	\$240
Normal economy	\$4	\$90
Recession	\$0	\$0

The company goes out of business if a recession hits. Calculate the expected rate of return and standard deviation of return to Leaning Tower of Pita shareholder. Assume for simplicity that the three possible states of the economy are equally likely. The stock is selling today for \$80.

- 2) The common stock of Escapist Films sells for \$25 a share and offers the following payoffs next year: (BBM 11.PP16)

State	Dividend	Stock Price
Boom	\$0	\$18
Normal economy	\$1	\$26
Recession	\$3	\$34

- a) Calculate the expected return and standard deviation of Escapist. All three scenarios are equally likely.
- b) Then calculate the expected return and standard deviation of a portfolio half invested in Escapist and half in Leaning Tower of Pita (Q1). Explain how expected return and standard deviation of the portfolio is related to the separate returns and standard deviations of both shares.
- c) Show the risk-return combinations of Leaning Tower of Pita, Escapist Films, and the portfolio in a diagram. Explain how the diversification effect can be seen in the graph.

Beta

- 3) A stock's beta measures the: (BMM 12.TB34)
- a. average return on the stock.
 - b. variability in the stock's returns compared to that of the market portfolio.
 - c. difference between the return on the stock and return on the market portfolio.
 - d. market risk premium on the stock.
- 4) When the overall market experiences a decline of 8%, an investor with a portfolio of aggressive stocks will probably experience: (BMM 12.TB36)
- a. portfolio losses of less than 8%.
 - b. portfolio losses greater than 8%.
 - c. portfolio gains of less than 8%.
 - d. portfolio gains greater than 8%.
- 5) The average of beta values for all individual stocks is: (BMM 12.TB38)
- a. greater than 1.0; most stocks are aggressive.
 - b. less than 1.0; most stocks are defensive.
 - c. unknown; betas are continually changing.
 - d. exactly 1.0; these stocks represent the market.
- 6) If a stock consistently goes down (up) by 1.6% when the market portfolio goes down (up) by 1.2%, then its beta: (BMM 12.TB40)
- a. equals 1.04.
 - b. equals 1.24.
 - c. equals 1.33.
 - d. equals 1.40.

- 7) Below you find a spreadsheet to calculate the beta of the Ford share using monthly data for six months. (Note: This is done for simplification purposes. Usually, when working with monthly data, you would include something like five years of data.) (BMM 12_spreadsheet question)

Month	Returns, percent		Formula used in Column C
	S&P 500	Ford	
Sep-10	5,74	4,43	
Aug-10	-4,74	-11,59	
Jul-10	6,88	26,69	
Jun-10	-5,39	-14,07	
May-10	-8,20	-9,91	
Apr-10	1,48	3,58	
Standard deviation (monthly)	5,75	13,99	=STDEVP(C3:C8)
Standard deviation, annualized	19,91	48,47	=C10*SQRT(12)
Beta		2,15	=SLOPE(C3:C8,B3:B8)
Correlation		0,88	=CORREL(C3:C8,B3:B8)

- Suppose that Ford's return in June 2010 had been -4% and its return in July 2010 had been 15%. Would you expect its beta to be more or less than the value obtained in the spreadsheet?
- Suppose that Ford's return in each month had been 1 percentage point higher than the values presented in the accompanying spreadsheet. Would Ford's beta differ from the value obtained in the spreadsheet?
- Suppose that you add one month of data to your spreadsheet and find that in October 2010, Ford was down 5% while the market was up 5%. Would you expect Ford's beta to be more or less than the value obtained in the spreadsheet?

Total Risk and Market Risk

- 8) What two elements are represented in security returns? (BMM 12.TB31)
- A premium for market risk and for unique risk
 - A premium for unique risk and a premium for firm-specific risk
 - A premium for diversification and a premium for portfolio risk
 - A premium for time value of money and a premium for market risk

- 9) What is the most logical explanation for a +2.0% return on a stock with a beta of 1.0 in a month where the market returned +1.0%? (BMM 12.TB43)
- a. The stock is aggressive.
 - b. The market is undervalued.
 - c. Favorable firm-specific news was reported.
 - d. The beta is incorrect.
- 10) A considerable scattering in the plot of points representing the historic returns of a stock versus the returns on the market indicates the: (BMM 12.TB52)
- a. high beta of the stock.
 - b. unique risk of the stock.
 - c. changes in market risk premium over time.
 - d. current underpricing of the stock.
- 11) Which of the following portfolios might be expected to exhibit less unique risk? (BMM 12.TB95)
- a. Five random stocks; portfolio beta = .8
 - b. Three random stocks; portfolio beta = 1.2
 - c. Ten random stocks; portfolio beta = 1.0
 - d. Twelve random stocks; portfolio beta unknown
- 12) The stock of Newmont Mining, the world's largest gold producer, has above-average volatility but relatively low beta. Why? (BMM 12.TB117)

Portfolio Beta

- 13) What is the beta of a three-stock portfolio including 25% of stock A with a beta of .90, 40% of stock B with a beta of 1.05, and 35% of stock C with a beta of 1.73? (BMM 12.TB47)

SML and CAPM

- 14) If Treasury bills are yielding 10% at a time when the market risk premium is 6%, then the: (BMM 12.TB54)
- a. market portfolio should yield 4%.
 - b. market portfolio should yield 6%.
 - c. market portfolio should yield 16%.
 - d. market portfolio should yield 22%.
- 15) What rate of return should an investor expect for a stock that has a beta of 1.25 when the market is expected to yield 14% and Treasury bills offer 6%? (BMM 12.TB64)
- 16) Why do stock market investors appear not to be concerned with unique risks when calculating expected rates of return? (BMM 12.TB65)
- a. There is no method to quantify unique risks.
 - b. Unique risks are assumed to be diversified away.
 - c. Unique risks are compensated by the risk-free rate.
 - d. Beta includes a component to compensate unique risk.
- 17) Calculate the expected rate of return for the following portfolio, based on a Treasury bill yield of 4% and an expected market return of 13%: (BMM 12.TB118)

Stock	Percent of Portfolio	Beta
A	20%	1.6
B	25%	1.2
C	10%	1.0
D	30%	0.9
E	15%	0.8

Exercises Chapter 6: Capital Market Based Cost of Capital**Basic Exercises****Applying the CAPM in Capital Budgeting**

- 1) A proposed investment must earn at least as much as the _____ if it is to be deemed acceptable. (BMM 12.TB82)
- company cost of capital
 - risk-free rate
 - market risk premium
 - project cost of capital
- 2) The project cost of capital is: (BMM 12.TB105)
- equal to the company cost of capital.
 - less than the company cost of capital.
 - greater than the company cost of capital.
 - not necessarily related to the company cost of capital.
- 3) Draw the SML when the Treasury bill rate is 4% and the market risk premium is 7%. Which of the following capital investments have positive NPVs? (BMM 12.PP12a)

Project	Beta	Internal Rate of Return, %
P	1.0	14
Q	0	6
R	2.0	18
S	0.4	7
T	1.6	20

Required Rates of Return of Single Components

- 4) According to CAPM estimates, what is the cost of equity for a firm with beta of 1.5 when the risk-free interest rate is 6% and the expected return on the market portfolio is 15%? (BMM 13.86)

- 5) What return on equity do investors seem to expect for a firm with a \$55 share price, an expected dividend of \$5.50, a beta of .9, and a constant growth rate of 5.5%? (BMM 13.87)
- 6) The Isabelle Corporation rents prom dresses in its stores across the southern United States. It has just issued a five-year, zero-coupon corporate bond (face value \$100) at a price of \$77. What is the cost of debt of this bond? (BDM 6.33)

Weighted-Average Cost of Capital (WACC)

- 7) The company cost of capital for a firm with a 65/35 debt/equity split, 8% cost of debt, and 15% cost of equity would be? (BMM 13.32)
- 8) The weighted-average cost of capital, after tax, for a firm with a 65/35 debt/equity split, 8% cost of debt, 15% cost of equity, and a 35% tax rate would be? (BMM 13.33)
- 9) Consider the following thought experiment: Your finance course runs its own little company (HSBA Finance GmbH). Your business portfolio has a beta of 1.2. You calculate with a WACC of 10%. You are quite happy as you received an incredible donation of EUR 1 million for your business, i.e., you get the money for free!

You consider using the EUR 1 million for a new attractive project which has the same risk profile as the HSBA Finance GmbH. What is your opportunity cost of capital for this project? 0%? Explain.